

AIMLABS 3D

Python · First-Person Aim Trainer · Real-Time 3D Rendering

OPENGL 3.3

PYTHON 3.10+

MODERNGL

SCORE 0000

ACC 0.0%

LEVEL Easy

0:34

Field Mode

HUMANOID TARGETS · BODY + HEAD HITBOXES

SCORE 0000

ACC 0.0%

LEVEL Easy

0:56

Arena Mode

SPHERE TARGETS · PHONG SHADING

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● COMPUTER GRAPHICS CONCEPTS

Phong Illumination Model	ambient·diffuse·specular
Vertex & Fragment Shaders (GLSL)	GPU pipeline
Model · View · Projection Matrices	MVP transform
VAO / VBO / IBO GPU Buffers	mesh upload
Depth Buffering & Depth-Disable	occlusion
Procedural Skybox (time uniform)	star shimmer
Analytic Ray-Sphere Intersection	raycasting
Surface-to-GL Texture Upload	HUD render
View-Space Gun Rendering	FPS camera
Normal Transform (Inverse Transpose)	lighting math
Pixel-Space Quad for HUD Overlays	2D in 3D
Procedural Sphere / Humanoid Geometry	mesh gen

● TECHNOLOGY STACK

ModernGL	OpenGL context, shaders, VAO/VBO, draw calls
Pygame	Window, input, audio, font surfaces
NumPy	Matrix buffers, procedural audio synthesis
Pyrr	View / projection / model matrix creation
GLSL	Custom shaders: phong · skybox · hud

● RUNTIME CONTROLS

L Click	Shoot / start	Mouse	Look / aim
WASD	Move	Shift	Sprint
M	Toggle mode	R	Reload
ESC	Pause / quit	R (end)	Restart

● GLSL SHADERS

phong.vert / .frag

World geometry, targets, gun. Phong lighting + hit_flash + stripe_mode uniforms.

skybox.vert / .frag

Full-screen backdrop, depth disabled. time uniform → star shimmer.

hud.vert / .frag

Text, crosshair, flash, vignette. Toggles use_texture uniform.

● RENDER PIPELINE

- 1 Clear color + depth
- 2 Skybox depth off
- 3 World panels
- 4 Obstacles
- 5 Targets
- 6 Gun (view-space) depth off
- 7 HUD overlays depth off

● AUDIO SYSTEM

Shot & reload from `game/sounds/`. Hit ping is procedurally synthesised via NumPy. Fallback when files unavailable.

● QUICK START

```
python -m venv .venv
pip install -r requirements.txt
python main.py
```